B.Tech. Degree VI Semester Special Supplementary Examination November 2012

ME 604 HEAT AND MASS TRANSFER

(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

(Approved data book may be permitted)

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) Show that the temperature profile for the heat conduction through a wall of constant thermal conductivity is a straight line.

(b) Explain Fourier law of heat conduction.

(c) Describe the hydrodynamic boundary layer.

(d) What is Reynold's analogy? Describe the relation between fluid friction and heat transfer.

(e) Distinguish between black body and gray body.

(f) State and explain Kirchoff's identity.

(g) Explain the different types of heat exchangers.

(h) What do you mean by fouling factor? Write the equations for overall heat transfer coefficient for a tube by considering fouling based on inside and outside diameter.

PART B

(4 × 15 = 60)

II. (a) Derive the temperature distribution and heat flow of a rectangular plate long fin of uniform cross section. (10)

(b) Calculate the heat loss per square metre surface area of a 40cm thick furnace having surface temperatures of 300°C and 50°C if the thermal conductivity of the material is $K = 0.005T - 5 \times 10^{-6} T^2$, where T is temperature in °C. (5)

OR

III. (a) What is meant by critical thickness of insulation? Derive the expression of critical radius of insulation of a hollow sphere. (7)

(b) An aluminium sphere weighing 5.5Kg and initially at a temperature of 290°C is suddenly immersed in a fluid at 15°C. The convective heat transfer coefficient is 58 W/m^2K. Estimate the time required to cool the aluminium to 95°C using the lumped capacity method analysis. (8)

IV. (a) Explain the significance of Reynold's number, Nusselt number and Prandtl number. (9)

(b) A fan provides air speed upto 50m/s is used in low speed wind tunnel with atmospheric air at 27°C. If this wind tunnel is used to study the boundary layer behaviour over a flat plate upto Re = 10^8, what should be the minimum plate length? At what distance from the leading edge would transition occur, if the critical Reynolds number $Re_{cr} = 5 \times 10^5$? (6)

OR

V. (a) Explain pool boiling and the different regimes of pool boiling with suitable sketches. (8)

(b) A metal plate 0.609 m high forms the vertical wall of an oven and is at temperature of 161°C. Within the oven air is at temperature of 93°C and one atmosphere. Assuming that the natural convection condition hold near the plate, estimate the mean heat transfer coefficient and the rate of heat transfer per unit width of the plate. (7)

VI. (a) How is Stefan-Boltzman law derived from Planck's law of thermal radiation? State the law also. (9)

(b) Define absorptivity, reflectivity and transmissivity. (6)

OR

VII. (a) Explain the radiation shield. (7)

(b) A pipe carrying steam having an outside diameter of 20cm runs in a large room and is exposed to air at a temperature of 30°C. The pipe surface temperature is 400°C. Calculate the loss of heat to surroundings per metre length of pipe due to thermal radiation. The emissivity of the pipe surface is 0.8. What would be the loss of heat due to radiation if the pipe is enclosed in a 40cm diameter brick conduit of emissivity 0.91? (8)

VIII. (a) Derive the LMTD for a parallel flow heat exchanger and also list the assumptions. (12)

(b) Define overall heat transfer coefficient. (3)

OR

IX. (a) In a double pipe counter flow heat exchanger, 10,000 Kg/hr of an oil having specific heat of 2095 J/KgK is cooled from 80°C to 50°C by 8000 Kg/hr of water entering at 25°C. Determine the exchanger area for an overall heat transfer coefficient of 300W/^{2}K. Take C_p of water as 4180 J/KgK. (12)

(b) What do you mean by NTU? (3)

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